

Process Analysis

Due: 2025-11-08 @ 2200**Weight:** 10 %**Submitter:** Individual

Assignment Task

In your process analysis, your task is as follows:

Choose two key observations that have been important for your learning during the design process thus far. Using a different Praxis concept/tool/model/framework (CTMF) for each observation, explain how each observation has influenced your developing understanding of the engineering design process. The observations can be based on specific individual actions or your team's design work overall.

Process Analysis in Context

In Praxis I, you will write two analyses of your experience with the engineering design process.¹ Each analysis challenges you to develop your self-awareness about how you practice engineering design, communication, and teamwork. This is part of *reflective practice*,² which is critical for developing as an engineer and engineering designer. Your ability to analyze and reflect on your experience enables you to gain more conscious control of your learning and practice. This will help you make more informed choices about how you approach engineering design activities, communicate, work in a team, and use the concepts, tools, models, and frameworks introduced throughout the course.

Each strand in our FDCR model focuses on one aspect of the design process and its associated activities. We teach framing, diverging, and converging in sequence in Praxis I, with representing being woven throughout. As you gain more experience with FDCR, you will notice that all of the strands influence each other most of the time. Examining your design process partway through your Praxis I project and again at the end of the course will support your developing understanding of the process. By putting the strands together, you will have a basis for your future design work and, more practically for one of your Praxis II assignments, your Engineering Design Portfolio. The process analyses also provide a record of your activities and growth.

We encourage you to pay attention to your teamwork and the engineering design process in lecture and studio. We encourage you to document and gather evidence of your process, as you will need it to support the claims you make in your process analysis.

¹ The first analysis happens partway through the design process, when you've gained experience with framing and diverging, and the second analysis will be incorporated into the final exam, when you have also gained experience with converging. You gain experience with representing throughout the design process.

² A book that can serve as a useful handbook for reflective practice is the following:
B. Bassot, *The reflective practice guide: An interdisciplinary approach to critical reflection*. London and New York: Routledge, 2016. This book is available through the U of T library at [this link](#).

Assignment Stakeholders

There are three main stakeholder groups for this assignment:

- You, individually, an aspiring engineer and team member, who will practice engineering design with different teams, in different contexts and in pursuit of differing goals, and so need to become aware of how you practice engineering design and communicate that practice.
- Your teaching team, who want you to become aware of how individual and team actions influence your team and your team's design work and to develop your own understanding of the engineering design process, and who have about twenty (≤ 20) minutes to evaluate your submission.
- Your current and future teammates, who will benefit if you develop better self-awareness and are able to use critical reflection to improve your learning and practice.

Writing the Assignment

A key part of reflective practice is engaging in critical reflection on your experiences, so that you can make the most of them and use them to inform future actions. To help you build your ability to engage in critical reflection, we recommend using the “What, so what, now what” model.³ Use the following tasks and questions to guide your work:

1. **What:** Choose **two (2)** “whats.” Consider how you and your team have work(ed) together thus far and observe how your team has used concepts, tools, models, and/or frameworks (CTMFs) in your process. Look at how the actions of individuals within the team and the team's actions as a whole affect your team's design process. Choose two observations from your process so far and briefly describe them. Make sure you include carefully selected evidence from your documentation of the team's activities to support claims you make about your observations.
2. **So what:** Analyze the “so what.” Analyze each observation to determine how it impacted your team activity. In your analysis, you should connect each observation to a distinct CTMF we use in Praxis. Though they may be related, you should use different CTMFs for each observation. What were the outcomes or consequences of the observation on your design process? How has this affected your understanding of the engineering design process thus far? What have you learned through analyzing each observation using your chosen CTMFs?
3. **Now what:** Consider the “now what.” Based on your analysis, briefly explain how you intend to maintain, change, or improve your **own** actions moving forward in your current design work and/or in future design work. Be specific about your strategy.

Note: This model sets up the guiding questions for this assignment, but is **not** intended to be a template for the structure of your analysis. You should use a structure that works best for *your* analysis. Don't answer all these questions like a problem set.

³ A helpful explanation of this model is available from Simple Psychology online at [What? So What? Now What? Critical Reflection Model](#).

Assignment Expectations

The key words “must”, “must not”, “required”, “should”, “should not”, or “recommended”, in this and following sections are to be interpreted as described in RFC 2119.

Objectives

1. Analyze and critically reflect on your experiences and learning in terms of:
 - insights you have gained about the behaviours, concepts, tools, models, and/or frameworks involved in the actions/observations you have analyzed
 - the implications of specific observations on your team’s design work
 - how selected Praxis CTMFs help you interpret and make meaning of specific observations and their significance for your learning
 - how you understand/interpret the engineering design process thus far
2. Document both behaviours and CTMFs you have used in your design work thus far.
3. Demonstrate clear, concise engineering communication.

Requirements

Your process analysis

1. **Shall** analyze two (2) specific observations, each using a distinct CTMF.
2. **Should** be between 700 and 1000 words.
3. **Shall** not exceed three (3) pages of integrated text and graphics.⁴
4. **Shall** include your name and the total word count at the top of the page.
5. **Shall** be formatted with 11-point font, 1.1-line spacing, and one-inch margins on standard letter-sized paper.
6. **Shall** be submitted as a single PDF file through Quercus to the correct assignment.
7. **Shall** have a file name that follows the Praxis naming convention:
<UTORID>-PA.pdf, e.g., lofgreen-PA.pdf

Evaluation Criteria

These criteria describe the attributes that will be evaluated. For the full metrics, consult the Process Analysis Independent Assessment Tool. That tool places these characteristics alongside descriptions of levels of achievement that you can use to independently assess the quality of your work. These criteria may not be weighted equally.

1. Quality of introduction to frame the process analysis
2. Quality of assessment of how specific observations, as analyzed using selected Praxis CTMFs, contribute to **your** understanding of the engineering design process thus far
3. Clarity of future action(s) based on analysis
4. Integration of relevant and clear evidence to support the analysis
5. Clarity of written and visual expression

⁴ If warranted, you may include an appendix, which will not count in the page limit. Appendix material is considered supplemental only (i.e., it may not get read during assessment), so it shall not include anything that is essential for your analysis. All essential content must be included in the main text.